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1. ABSTRACT

Online shopping platforms depend heavily on customer feedback to understand product quality and delivery problems. A review may describe an ordinary product concern, but it may also point to a possible loss event—for example, an item missing from a parcel, damaged packaging, a suspected substitution, or a package that appears to have been opened. When feedback arrives in large quantities, finding these signals one review at a time becomes difficult.

This project develops a machine-learning backend that screens e-commerce reviews for unusual patterns that could be relevant to retail shrinkage. It combines information from the review wording with selected numerical indicators. The review heading and review body are joined and converted to lowercase. TF-IDF represents the text numerically, while a rating–sentiment difference and a keyword indicator provide additional signals. PCA compresses the text representation into two components. Isolation Forest is then used to flag unusual records, and K-Means separates the flagged records into three groups.

The implementation has a model-training workflow and a separate prediction workflow. During training, the transformations and estimators are fitted and saved. During prediction, the saved objects are loaded and applied to incoming reviews in the same sequence. The output includes an anomaly flag, a decision score, and a risk-cluster assignment. These outputs are intended to help staff decide which reviews deserve closer examination; they should not be interpreted as confirmation that shrinkage or fraud has occurred.

2. INTRODUCTION AND PROBLEM STATEMENT

2.1 Introduction

A retail business can lose goods for several reasons, including theft, damage, missing contents, and problems involving returned packages. In online retail, the business often receives descriptions of these events through customer reviews and complaints. Such descriptions are written in everyday language and may be mixed with comments about price, product quality, delivery speed, or customer service.

The challenge is not simply to search for a few words. A review that mentions a missing item may be important, while the same word in another context may not indicate a loss at all. A useful screening process therefore needs to consider more than one signal. This project applies text processing and unsupervised machine learning to identify reviews that differ from the usual patterns in the available data.

2.2 Problem Statement

The project addresses the difficulty of locating potentially shrinkage-related reviews within a large collection of e-commerce feedback. Reading every record manually requires time and may lead to relevant cases being missed. The proposed backend processes review text and selected review attributes, marks unusual records, and organizes the marked records into groups that can guide further inspection.

2.3 Objectives

- Prepare a consistent text field from each review title and review body.
- Derive numerical indicators from the star rating, sentiment category, and shrinkage-related vocabulary.
- Convert review text into TF-IDF features and reduce the text feature space with PCA.
- Apply Isolation Forest to identify records that appear unusual in the engineered feature space.
- Use K-Means to organize detected anomalies into three broad risk patterns.
- Save the fitted preprocessing components and models for reuse when new reviews are received.

3. RELATED WORK / BACKGROUND

The technical approach is based on established methods in text representation, dimensionality reduction, anomaly detection, and clustering. The project documentation specifies the algorithms used in the implementation but does not provide a review of named research papers or a comparison with other systems.

This section therefore explains the technical ideas that form the basis of the implemented pipeline.

3.1 Text representation with TF-IDF

Machine-learning estimators generally require numerical input, whereas customer feedback is written as text. TF-IDF provides a way to represent documents using the importance of words within a document and across the document collection. The project uses a TF-IDF vectorizer with a maximum of 500 features, limiting the representation to the selected vocabulary.

3.2 Principal Component Analysis

Text vectorization can create a feature space with many dimensions. PCA transforms the original features into a smaller set of components. In the supplied implementation, the TF-IDF representation is reduced to two components, referred to as Text_PCA1 and Text_PCA2. These components are used as part of the numerical representation supplied to the later stages.

3.3 Isolation Forest

Isolation Forest is an unsupervised method for finding observations that are relatively easy to separate from the rest of a dataset. It does not require a class label for every training example. The project sets its contamination parameter to 0.05. The model returns an anomaly indicator and a decision score; according to the project documentation, a lower decision score represents a more severe anomaly.

3.4 K-Means

K-Means divides observations into a specified number of groups based on their positions in the feature space. Here it is used after anomaly detection, rather than as the first screening step. The configured value is K = 3, and the resulting groups are given operational descriptions based on the patterns associated with each cluster.

4. DATASET DESCRIPTION

The input to the system is a collection of e-commerce customer reviews. The backend documentation identifies review title, review text, star rating, and quality-sentiment information as inputs or derived fields used by the processing steps. It does not state the original dataset's exact name, download location, record count, collection dates, or complete schema. Those particulars should be taken from the dataset actually used by the project and inserted before submission.

4.1 Main fields and their use

Field or feature	How it is used
Review_Title	The short heading written by the reviewer; combined with the main review text.
Review_Text	The main written feedback supplied by the customer.
Full_Review	The title and review text joined together and converted to lowercase.
Star_Rating	The customer's star score, divided by 5.0 to place it on a 0–1 scale.
Quality_Sentiment_Num	Numerical sentiment value: Positive = 1.0, Neutral = 0.5, Negative = 0.0.
Rating_Quality_Mismatch	Absolute difference between the normalized star rating and the numerical sentiment value.
Shrinkage keyword flag	A text-derived indicator showing whether selected shrinkage-related terms occur in the review.

4.2 Preparation of the input records

The title and review body are brought together so that the text-processing stage can examine the complete written comment. The combined text is lowercased to make matching less dependent on capitalization. The rating is normalized by dividing it by five, and the sentiment category is converted to its assigned numerical value. The mismatch feature is calculated from these two values. A regular-expression check also looks for words connected with missing contents, theft, packaging interference, or product condition.

No dataset size, missing-value summary, class balance, or train/test partition is specified in the supplied backend document. Such values are intentionally not reported here rather than being estimated.

5. METHODOLOGY (DATA CLEANING, ETL, MODEL BUILDING)

5.1 Pipeline structure

The implementation is separated into a training workflow and a prediction workflow. The training workflow is described in pipeline.ipynb. It reads the review data, prepares the text and numerical features, fits the transformations, trains the estimators, and saves the fitted objects. The prediction workflow is described in predict.ipynb. It reloads the saved objects and applies the same preparation steps to new records before producing outputs.

5.2 Cleaning and preparing review text

- Join Review_Title and Review_Text, with a space between them, to form Full_Review.
- Convert Full_Review to lowercase so that text handling is consistent.
- Convert the star rating to a normalized value by dividing Star_Rating by 5.0.
- Map Positive, Neutral, and Negative quality sentiment to 1.0, 0.5, and 0.0 respectively.
- Check the combined text for the shrinkage-related terms specified in the backend documentation.

5.3 Feature engineering

One feature measures the difference between the customer's numerical rating and the mapped sentiment category. It is calculated using the absolute difference shown below:

$$\textbf{Rating_Quality_Mismatch} = \left| \left(\frac{\textbf{Star_Rating}}{5.0}\right) - \textbf{Quality_Sentiment_Num} \right|$$

A high value means that the written sentiment category and the normalized star rating are far apart. This is a signal for review, not a standalone finding of suspicious activity. The keyword scanner checks for the terms empty, missing, stolen, tampered, unsealed, swapped, used, fake, broken, and tape. A match adds a further indicator that the text may relate to a logistics or product-loss issue.

5.4 TF-IDF and PCA processing

After text preparation, the TF-IDF vectorizer converts Full_Review into a numerical document representation. The configured vocabulary is limited to the top 500 features. PCA is then applied to the vectorized text and reduces it to two values for each record: Text_PCA1 and Text_PCA2. The reduced text features are used alongside the other engineered values.

5.5 Scaling numerical features

The pipeline uses StandardScaler to scale the numerical input features before model use. The fitted scaler is saved so that the prediction workflow can apply the same scaling parameters to incoming records. Reusing the training-time scaler is important; fitting a different scaler on each new batch could make the feature values inconsistent with those seen by the trained model.

5.6 Model construction

First stage – Isolation Forest: The model examines the prepared feature space and identifies records that appear different from the majority. Its contamination parameter is configured as 0.05. The output includes Is_Shrinkage_Anomaly and Anomaly_Score. Lower decision scores are treated as indicating greater anomaly severity in this project.

Second stage – K-Means: The reviews identified as anomalies are further grouped using K-Means with three clusters. The cluster descriptions in the project documentation are Value/Quality Dissatisfaction, Rating Mismatch / Suspected Fake, and Standard Product Return Risk.

5.7 Saving the trained components and making predictions

The fitted preprocessing objects and estimators are saved as joblib files in the models/ directory. The prediction workflow loads the scaler, TF-IDF vectorizer, PCA transformer, Isolation Forest estimator, and K-Means estimator. New reviews are transformed using the saved vectorizer and PCA transformer; numerical features are scaled using the saved scaler; the trained models then generate the anomaly and cluster outputs.

Saved file	Stored object / role
models/scaler.pkl	Fitted StandardScaler used for feature scaling.
models/tfidf_vectorizer.pkl	Fitted TF-IDF vectorizer and its learned vocabulary.
models/pca_transformer.pkl	Fitted PCA transformation for text features.
models/isolation_forest_model.pkl	Trained anomaly-detection estimator.
models/kmeans_cluster_model.pkl	Trained three-cluster K-Means estimator.

The backend documentation identifies REST API integration, for example through FastAPI, as a possible way for another application to submit reviews and receive predictions. It also mentions a simple Flask prediction application. The document does not specify a production deployment, hosting environment, or completed API contract.

6. RESULTS AND EVALUATION

The documented system produces three main kinds of output: an anomaly indicator, an anomaly decision score, and a cluster assignment for a flagged review. The Isolation Forest configuration uses a contamination value of 0.05, and the K-Means stage uses three clusters. The cluster descriptions below follow the supplied backend documentation.

6.1 Interpretation of the three clusters

Cluster index	Project label	Description in operational terms
0	Value/Quality Dissatisfaction	Reviews with substantial complaint content relating to defects, poor quality, or

		expectations that were not met.
1	Rating Mismatch / Suspected Fake	Reviews showing a strong difference between the star rating and the sentiment expressed in the text.
2	Standard Product Return Risk	Reviews with frequent references to missing contents, broken packaging, stolen goods, or tampered sealing.

6.2 Evaluation and what can be concluded

The available technical document records the model settings and explains the intended meaning of the model outputs. It does not include a labelled test set or measured values for precision, recall, F1-score, accuracy, or a confusion matrix. For that reason, this report does not assign a numerical performance score to the system.

The contamination setting is a parameter supplied to Isolation Forest; it is not an accuracy measurement and does not demonstrate that the flagged records are genuine shrinkage cases. Likewise, the names assigned to the K-Means groups are interpretations intended to make the output easier to review. Confirmation requires checking the original transaction, delivery, return, and customer-support evidence.

7. DISCUSSION

The design uses several kinds of evidence because no single review attribute is sufficient to describe a possible shrinkage incident. Text features retain information about the words used by the customer. The mismatch feature identifies cases where the star score and the mapped sentiment do not agree closely. The keyword flag focuses on vocabulary that may describe missing goods or package interference.

Together, these inputs give the anomaly detector a broader view of each review than a keyword search alone.

The two-stage model design also serves different purposes. Isolation Forest performs the initial screening and marks unusual records. K-Means then groups the marked records so that reviewers can distinguish broad patterns instead of receiving one undifferentiated list. For example, a group dominated by product-quality complaints may require a different response from a group containing many references to missing parcel contents.

The outputs still require careful interpretation. A customer might write the word “broken” while discussing a feature unrelated to packaging, and a mismatch between rating and sentiment can occur because of sarcasm or unclear language. An unusual review is not necessarily a fraudulent review. Similarly, cluster membership describes similarity to other records in the model's feature space; it does not establish the cause of the customer's experience.

The current documentation does not provide a labelled evaluation set or confirmed investigation outcomes. A practical next step is therefore to compare model flags with cases reviewed by staff. This would help determine whether the selected features and cluster descriptions are useful in the business process and where the pipeline needs refinement.

8. CONCLUSION AND FUTURE WORK

8.1 Conclusion

This project sets out a backend workflow for screening e-commerce customer reviews for patterns that may be relevant to retail shrinkage. It prepares review text, creates rating- and keyword-based features, represents text using TF-IDF, reduces the text dimensions with PCA, and scales the numerical input. Isolation Forest is used to flag unusual records, after which K-Means assigns the flagged reviews to one of three risk-pattern groups.

The saved model artifacts allow the prediction workflow to reuse the same fitted transformations and estimators when processing new reviews. In this way, the system can help organize customer feedback and direct attention towards records

that may warrant further checks. Its role is to support investigation, not to make a final judgement about fraud, theft, or responsibility.

8.2 Future Work

- Build a verified set of past cases so that model predictions can be compared with confirmed outcomes.
- Calculate precision, recall, and F1-score when reliable reference labels are available.
- Review the keyword list with operational staff and test whether it captures relevant wording without producing too many unrelated matches.
- Compare the current TF-IDF representation with other suitable text-representation methods.
- Assess the stability and usefulness of the three clusters using reviewed examples.
- Connect the prediction workflow to a REST API and a dashboard for case review.
- Monitor incoming data and periodically reassess the pipeline as customer language and retail processes change.

9. REFERENCES

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